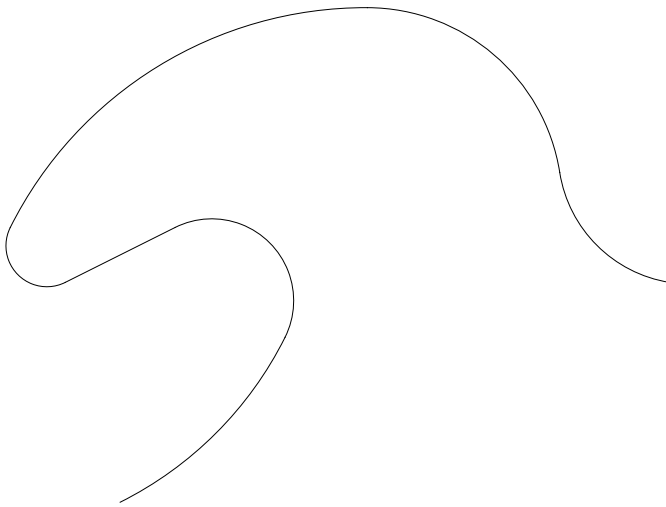


This notebook generates curves for the taxicab half-plane model project. It requires a circular-curves notebook to be run first.

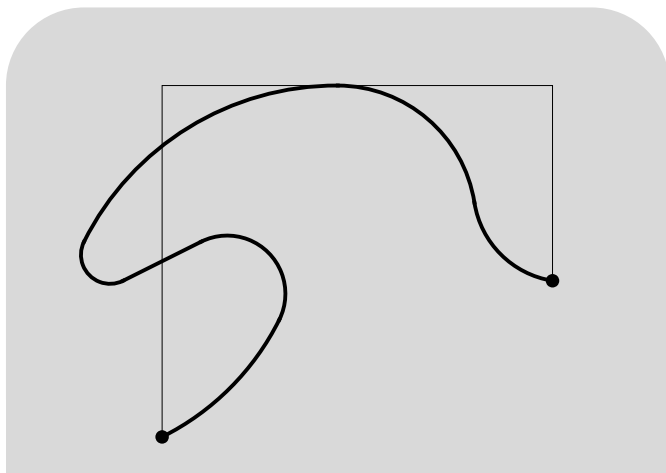
```
In[*]:= points = {{0, 1}, {3, 4}, {1, 6}, {-1, 5}, {-2, 6}, {4.5, 10}, {8, 7}, {10, 5}};
vectorlist = {{1, .5}};
adj = {{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}};
Do[vectorlist =
  Append[vectorlist, newerv[points[[i]], points[[i + 1]], vectorlist[[i]], adj[[i]]],
  {i, Length[points] - 2}]
arclist =
  Table[arc[points[[i]], points[[i + 1]], vectorlist[[i]], {i, Length[points] - 1}];
gamma = arclist;
Graphics[{arclist, (*Point[points]*)}]
```

Out[*]=



```
In[*]:= Graphics[{{LightGray, Rectangle[{-4, 0}, {13, 12}, RoundingRadius -> 2]},
  {LightGray, Rectangle[{-4, 0}, {13, 2}]}, {Dashed, Line[{-4, 0}, {13, 0}]}],
  Line[{{0, 1}, {0, 10}, {10, 10}, {10, 5}}], {Thick, gamma},
  {PointSize[Large], Point[{{0, 1}, {10, 5}}]}]
```

Out[*]=



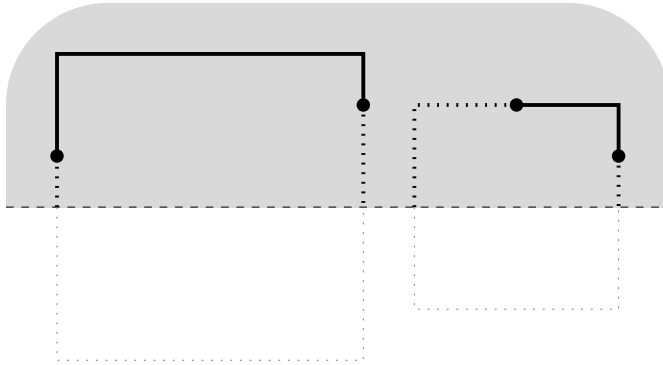
```
In[*]:= (*Export["gamma-and-eta.pdf", %132] *)
```

```

In[*]:= Graphics[{{LightGray, Rectangle[{-1, 0}, {12, 4}, RoundingRadius -> 2]},
  {LightGray, Rectangle[{-1, 0}, {12, 2}]}, {Dashed, Line[{{-1, 0}, {12, 0}}]},
  {Thick, Line[{{0, 1}, {0, 3}, {6, 3}, {6, 2}}]},
  {Thick, Dotted, Line[{{0, 0}, {0, 1}}, {{6, 2}, {6, 0}}]},
  {Gray, Dotted, Line[{{0, 0}, {0, -3}, {6, -3}, {6, 0}}]},
  {Thick, Line[{{9, 2}, {11, 2}, {11, 1}}]},
  {Thick, Dotted, Line[{{7, 0}, {7, 2}, {9, 2}}, {{11, 1}, {11, 0}}]},
  {Gray, Dotted, Line[{{7, 0}, {7, -2}, {11, -2}, {11, 0}}]},
  {PointSize[Large], Point[{{0, 1}, {6, 2}, {9, 2}, {11, 1}}]}}]

```

Out[*]=



```

In[*]:= (*Export["geodesics.pdf",%130]*)

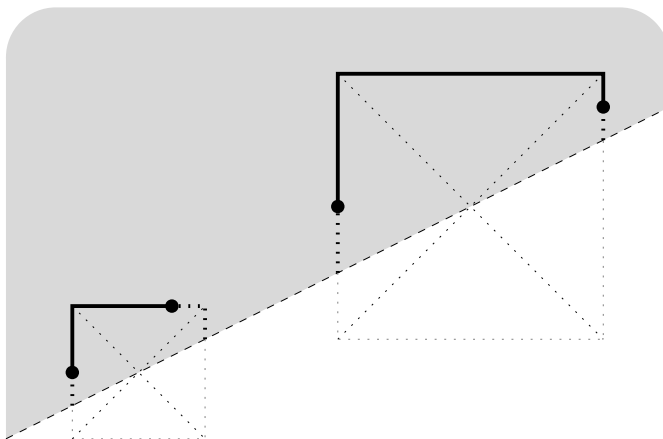
```

```

In[*]:= Graphics[{{LightGray, Rectangle[{-10, 5}, {10, 8}, RoundingRadius -> 1.5]},
  {LightGray, Polygon[{{-10, -5}, {10, 5}, {10, 6.5}, {-10, 6.5}}]},
  {Dashed, Line[{{-10, -5}, {10, 5}}]},
  {Thick, Line[{{0, 2}, {0, 6}, {8, 6}, {8, 5}}]},
  {Thick, Dotted, Line[{{0, 0}, {0, 2}}, {{8, 4}, {8, 5}}]},
  {Gray, Dotted, Line[{{0, 0}, {0, -2}, {8, -2}, {8, 4}}]},
  {Dotted, Line[{{0, 6}, {8, -2}}, {{0, -2}, {8, 6}}]},
  {Thick, Line[{{-8, -3}, {-8, -1}, {-5, -1}}]},
  {Thick, Dotted, Line[{{-4, -2}, {-4, -1}, {-5, -1}}, {{-8, -4}, {-8, -3}}]},
  {Gray, Dotted, Line[{{-4, -2}, {-4, -5}, {-8, -5}, {-8, -4}}]},
  {Dotted, Line[{{-8, -1}, {-4, -5}}, {{-8, -5}, {-4, -1}}]},
  {PointSize[Large], Point[{{0, 2}, {8, 5}, {-8, -3}, {-5, -1}}]}}]

```

Out[*]=

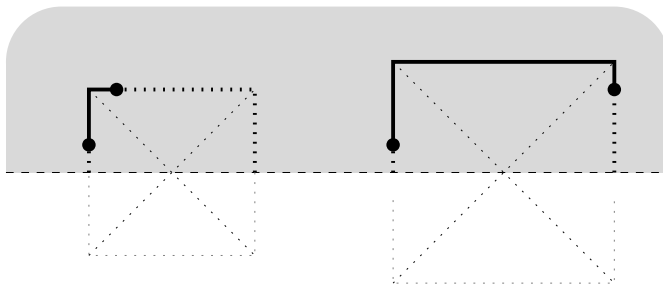


```

In[*]:= Graphics[{{LightGray, Rectangle[{-12, 0}, {12, 4}, RoundingRadius -> 2]},
  {LightGray, Polygon[{{-12, -2}, {12, -2}, {12, 2}, {-12, 2}]}},
  {Dashed, Line[{{-12, -2}, {12, -2}]}},
  {Thick, Line[{{2, -1}, {2, 2}, {10, 2}, {10, 1}]}},
  {Thick, Dotted, Line[{{2, -2}, {2, -1}}, {{10, -2}, {10, 1}]}},
  {Gray, Dotted, Line[{{2, -3}, {2, -6}, {10, -6}, {10, -3}]}},
  {Dotted, Line[{{2, 2}, {10, -6}}, {{2, -6}, {10, 2}]}},
  {Thick, Line[{{-9, -1}, {-9, 1}, {-8, 1}]}},
  {Thick, Dotted, Line[{{-3, -2}, {-3, 1}, {-8, 1}}, {{-9, -2}, {-9, -1}]}},
  {Gray, Dotted, Line[{{-3, -2}, {-3, -5}, {-9, -5}, {-9, -2}]}},
  {Dotted, Line[{{-9, -5}, {-3, 1}}, {{-9, 1}, {-3, -5}]}},
  {PointSize[Large], Point[{{2, -1}, {10, 1}, {-9, -1}, {-8, 1}]}]}]

```

Out[*]=



```

In[*]:= (*Export["geodesics0.pdf",%142]*)

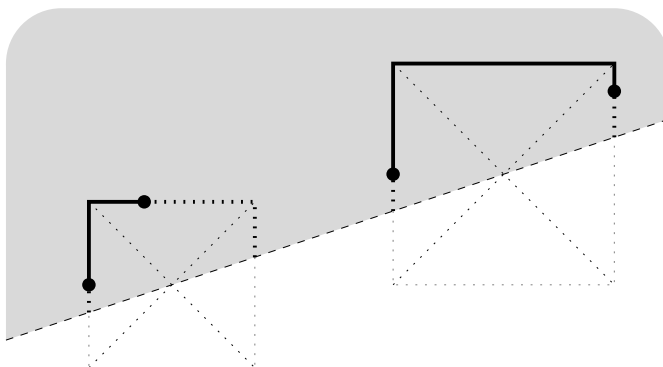
```

```

In[*]:= Graphics[{{LightGray, Rectangle[{-12, 4}, {12, 8}, RoundingRadius -> 2]},
  {LightGray, Polygon[{{-12, -4}, {12, 4}, {12, 6}, {-12, 6}]}},
  {Dashed, Line[{{-12, -4}, {12, 4}]}},
  {Thick, Line[{{2, 2}, {2, 6}, {10, 6}, {10, 5}]}},
  {Thick, Dotted, Line[{{2, 2/3}, {2, 2}}, {{10, 10/3}, {10, 5}]}},
  {Gray, Dotted, Line[{{2, 2/3}, {2, -2}, {10, -2}, {10, 10/3}]}},
  {Dotted, Line[{{2, 6}, {10, -2}}, {{2, -2}, {10, 6}]}},
  {Thick, Line[{{-9, -2}, {-9, 1}, {-7, 1}]}},
  {Thick, Dotted, Line[{{-3, -1}, {-3, 1}, {-7, 1}}, {{-9, -3}, {-9, -2}]}},
  {Gray, Dotted, Line[{{-3, -1}, {-3, -5}, {-9, -5}, {-9, -3}]}},
  {Dotted, Line[{{-9, -5}, {-3, 1}}, {{-9, 1}, {-3, -5}]}},
  {PointSize[Large], Point[{{2, 2}, {10, 5}, {-9, -2}, {-7, 1}]}]}]

```

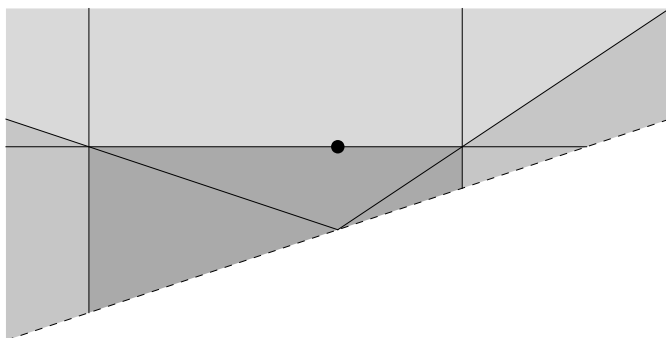
Out[*]=



```
In[*]:= (*Export["geodesics3.pdf",%100]*)
```

```
In[*]:= Graphics[{{LightGray, Polygon[{{-12, 4}, {-9, 3}, {9/2, 3}, {12, 8}, {-12, 8}}]},
  {Lighter[Gray], Polygon[{{-9, -3}, {9/2, 3/2}, {9/2, 3}, {-9, 3}}]},
  {Lighter[Lighter[Gray]], Polygon[{{-12, -4}, {-9, -3}, {-9, 3}, {-12, 4}}]},
  {Lighter[Lighter[Gray]], Polygon[{{9/2, 3/2}, {9/2, 3}, {12, 8}, {12, 4}}]},
  {Dashed, Line[{{-12, -4}, {12, 4}}]},
  {Line[{{-12, 3}, {9, 3}}, {{0, 0}, {12, 8}}, {{0, 0}, {-12, 4}}, {{9/2, 3/2},
    {9/2, 8}}, {{-9, -3}, {-9, 8}}]}], {PointSize[Large], Point[{{0, 3}}]}}
```

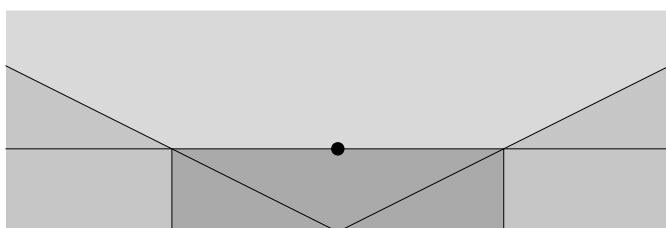
```
Out[*]=
```



```
In[*]:= (*Export["regions.pdf",%123]*)
```

```
In[*]:= Graphics[
  {{LightGray, Polygon[{{-12, 8}, {-12, 6}, {-6, 3}, {6, 3}, {12, 6}, {12, 8}}]},
  {Lighter[Gray], Polygon[{{-6, 0}, {6, 0}, {6, 3}, {-6, 3}}]},
  {Lighter[Lighter[Gray]], Polygon[{{-12, 6}, {-6, 3}, {-6, 0}, {-12, 0}}]},
  {Lighter[Lighter[Gray]], Polygon[{{6, 0}, {6, 3}, {12, 6}, {12, 0}}]},
  {Dashed, Line[{{-12, 0}, {12, 0}}]},
  {Line[{{-12, 3}, {12, 3}}, {{0, 0}, {12, 6}}, {{0, 0}, {-12, 6}}, {{6, 0},
    {6, 3}}, {{-6, -0}, {-6, 3}}]}], {PointSize[Large], Point[{{0, 3}}]}}
```

```
Out[*]=
```



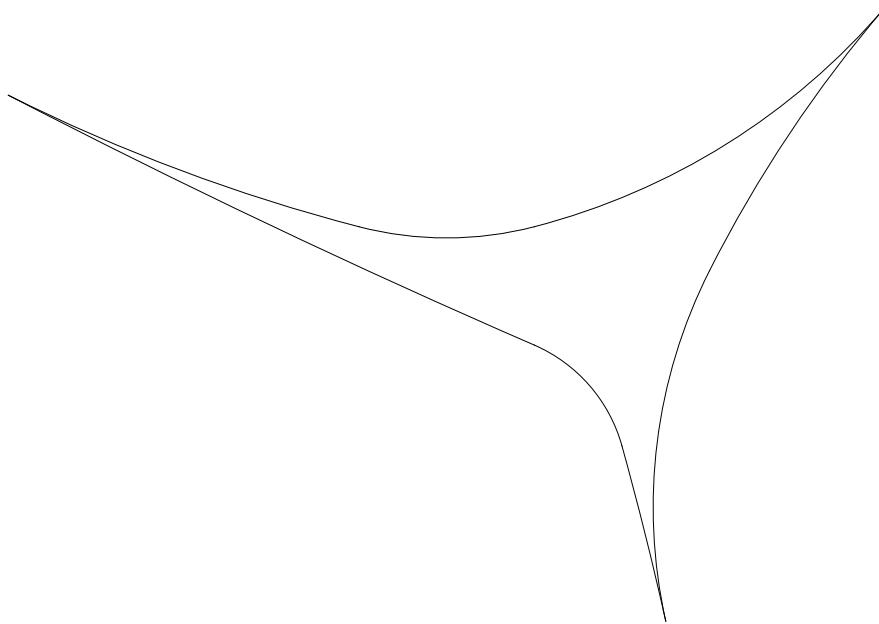
```
In[*]:= (*Export["regions0.pdf",%159]*)
```

```

In[ ]:= points =
  {{0, 9}, {4, 7.5}, {6, 7.5}, {10, 10}, {8, 7}, {7.5, 3}, {7, 5}, {6, 6.15}, {0, 9}};
vectorlist = {{2, -1}};
adj = {{0, 0}, {0, 0}, {0, Pi}, {0, 0}, {0, Pi}, {0, 0}, {0, 0}, {0, 0}};
Do[vectorlist =
  Append[vectorlist, newerv[points[[i]], points[[i + 1]], vectorlist[[i]], adj[[i]]],
  {i, Length[points] - 2}]
arclist =
  Table[arc[points[[i]], points[[i + 1]], vectorlist[[i]], {i, Length[points] - 1}];
triangle = arclist;
Graphics[{arclist(*,Point[points[*])}]

```

Out[]=

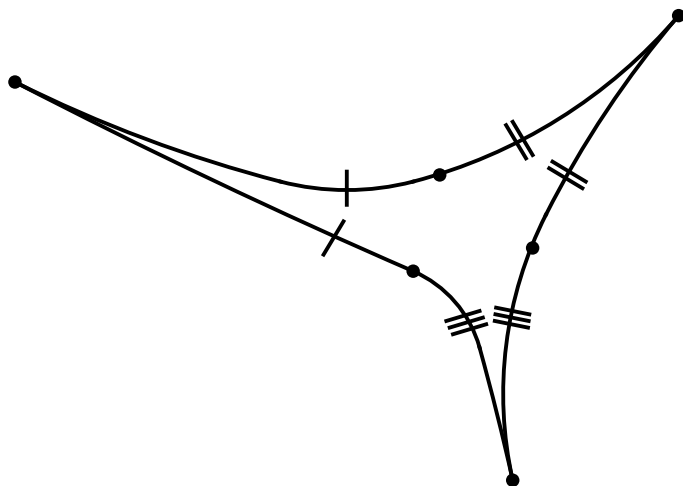


```

In[*]:= Graphics[{{Thick, triangle}, {PointSize[Large],
  Point[{{0, 9}, {10, 10}, {7.5, 3}, {6.4, 7.6}, {6, 6.15}, {7.8, 6.5}]}]},
  {Thick, Line[{{5, 7.15}, {5, 7.65}}, {{4.65, 6.4}, {4.95, 6.9}},
    {{7.4, 8.35}, {7.7, 7.85}}, {{7.5, 8.4}, {7.8, 7.9}},
    {{8.1, 7.8}, {8.6, 7.5}}, {{8.05, 7.7}, {8.55, 7.4}},
    {{6.5, 5.4}, {7., 5.55}}, {{6.55, 5.3}, {7.05, 5.45}},
    {{6.6, 5.2}, {7.1, 5.35}}, {{7.75, 5.5}, {7.25, 5.6}},
    {{7.74, 5.4}, {7.24, 5.5}}, {{7.73, 5.3}, {7.23, 5.4}}]}]}]

```

Out[*]=



```

In[*]:= (*Export["triangle.pdf",%540]*)

```